

Engineering Mathematics

Previous Year Questions • 1994–2026

COMPLEX VARIABLE ANALYSIS

(EC, EE, IN, CS, ME, CE, CH, PI)

Himanshu Agarwal & Anish Saha

PrepFusion™

Where Concept Meets Clarity!

Preface

Welcome to PrepFusion™!

This book is a complete compilation of the questions GATE has actually asked in Complex Variable Analysis (EC/EE/IN/CS/ME/CE/CH/PI), gathered from the original papers and arranged unit by unit and year by year. Nothing is paraphrased or reconstructed: every question appears as it was set, tagged with its year, session and marks.

Each question carries an identifier that maps one-to-one onto its worked solution in the companion solutions volume, so you can move between practice and explanation without a lookup table. Answer keys close every section, and answers revised by the examining authority (MTA) are marked where they occur.

Used end to end, the book shows you what the paper rewards — which topics repeat, how the framing has shifted over the years, and where your preparation still has gaps.

Meet your Authors

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> *"We built this book the way every aspirant deserves one to be built — organized strictly by unit and by year, so that every hour of practice maps directly onto how GATE actually tests you."*
>
> — Himanshu Agarwal & Anish Saha

How to Read the Question Numbers

Every question in this book carries a three-part identifier instead of a plain serial number:

Q3.24.7

3	Unit (chapter) number — Unit III of this book
24	GATE exam year — 2024 (two digits; 98 = 1998)
7	Question number within that unit and year

So **Q3.24.7** is the 7th question that GATE 2024 contributed to Unit III. Numbers are therefore not sequential through the book — they tell you the unit and the exam year at a glance. The same identifier is used in the questions book, the solutions book and the answer keys, so you can jump between them without a lookup table. In the PDF bookmarks panel, expand a unit to see every question ID and click one to go straight to it.

What the Bubbles Mean

Every question ends with three empty circles for you to shade in yourself:



 Easy  Medium  Hard

The companion solutions volume marks the same question with *our* recommended difficulty, using the identical bubbles. If you think a rating should be different, tell us directly — report it under the **Study Hub** section at **pyq.prepfusion.in**.

How to Read the Branch Snapshot

A Maths unit can span up to eight branches (EC, EE, IN, CS, ME, CE, CH, PI), each weighted very differently in the real exam. Right after every Unit divider, a **Branch Snapshot** page lists every branch that unit covers, in the same order the questions themselves follow, each in its own compact box:

Questions	the real total for that branch.
MCQ, MSQ, NAT	that same total, split by question type.
1M, 2M	how many of those questions carry 1 or 2 marks.
Descriptive	questions worth more than 2 marks, or whose marks value isn't recorded.
Avg Weightage	the average marks contributed per real GATE sitting, not a raw year-by-year total — a year GATE ran multiple real sessions (a genuine forenoon/afternoon split, not the same paper's Set A/B/C/D booklet variants) is divided across those sessions first, so a heavily multi-session year cannot silently outweigh a single-session one. The ~ marks it as a rounded figure, not an exact one.
Range	the lowest and highest that same per-sitting value has been, with the year each occurred.

Avg Weightage and Range are computed from **2010 onward** only: GATE dropped from a 150-mark to a 100-mark paper in 2009, but that year itself was a one-off transition (60 questions, not the 65-question format that became standard from 2010 on) — so 2010 is the first year fully comparable to modern papers, and marks-based figures from earlier years are left out of these two figures specifically (every other count on this page still includes every year). They also depend on knowing which GATE years ran more than one real sitting for a given branch — our own research into this is thorough for some branches and still a work in progress for others. Treat both figures as a **beta**, best-effort estimate: directionally useful, not exact.

MSQ questions specifically weren't introduced until 2021 — an older-year-heavy branch showing **MSQ: 0** reflects that real exam history, not a data gap.

Worked example

Branch Snapshot*

Branch: EC	**Questions:** 45	**MCQ:** 30	**MSQ:** 3	**NAT:** 12
1M: 20	**2M:** 22	**Descriptive:** 3	**Avg Weightage** ~4M	**Range:** 2M(2015)–6M(2021)

45 real EC questions on this unit; on average EC has drawn about 4 marks per real GATE sitting since 2010, ranging from as low as 2 (in 2015) to as high as 6 (in 2021).

A quick way to use this: a high **Avg Weightage** for your branch points to a consistently high-yield topic worth prioritizing; a low one is largely someone else's problem. A wide **Range** usually means the topic's importance has shifted over the years, so it is worth checking *which* years drove that before deciding how much weight to give it today.

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Scan any code with your phone camera, or tap the link beneath it. **pyq.prepfusion.in** is also where you can read this book's worked solutions online — and in its **Study Hub** section, report an error in any question or request a video solution for it.

Contributors

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UNIT

I

Complex Variable Analysis

Topics

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5. Milne Thompson and Total Derivative Method
6. Line Integration of Complex Variables
7. Points of Singularity and Laurent Series
8. Concept of Residue
9. Cauchy's Integral and Residue Theorem

Note to Reader

Complex Variable Analysis is present in branches with paper codes AE, BM, CH, EC, EE, IN, MA, MN, ME, NM, PE, XE according to GATE 2027 syllabus. Before starting please check with the latest syllabus.

Branches: EC,EE,IN,ME,CE,CH,PI

Branch Snapshots*

EC — Electronics & Communication Engineering

Questions: 31 **MCQ:** 22 **MSQ:** 1 **NAT:** 8
1M: 17 **2M:** 14 **Desc:** 0 **Avg Weightage** ~2M **Range:** 1M(2022)–2M(2025)

EE — Electrical Engineering

Questions: 27 **MCQ:** 23 **MSQ:** 1 **NAT:** 3
1M: 17 **2M:** 10 **Desc:** 0 **Avg Weightage** ~2M **Range:** 1M(2020)–3M(2021)

IN — Instrumentation Engineering

Questions: 30 **MCQ:** 23 **MSQ:** 3 **NAT:** 4
1M: 17 **2M:** 13 **Desc:** 0 **Avg Weightage** ~2M **Range:** 1M(2026)–4M(2022)

ME — Mechanical Engineering

Questions: 26 **MCQ:** 21 **MSQ:** 0 **NAT:** 5
1M: 12 **2M:** 14 **Desc:** 0 **Avg Weightage** ~2M **Range:** 0M(2015)–3M(2020)

CE — Civil Engineering

Questions: 8 **MCQ:** 8 **MSQ:** 0 **NAT:** 0
1M: 5 **2M:** 3 **Desc:** 0 **Avg Weightage** ~2M **Range:** 1M(2011)–1M(2011)

CH — Chemical Engineering

Questions: 14 **MCQ:** 12 **MSQ:** 0 **NAT:** 2
1M: 8 **2M:** 6 **Desc:** 0 **Avg Weightage** ~2M **Range:** 1M(2025)–3M(2015)

PI — Production & Industrial Engineering

Questions: 15 **MCQ:** 13 **MSQ:** 1 **NAT:** 1
1M: 11 **2M:** 4 **Desc:** 0 **Avg Weightage** ~2M **Range:** 1M(2026)–3M(2025)

**See preface for how Avg Weightage and Range are calculated.*

Electronics & Communication Engineering

Q5.25.1 MSQ 2025 2M

Which of the following statements involving contour integrals (evaluated counter-clockwise) on the unit circle C in the complex plane is/are TRUE?

- **A.** $\oint_C e^z\, dz = 0$
- **B.** $\oint_C z^n\, dz = 0$, where n is an even integer
- **C.** $\oint_C \cos z\, dz \neq 0$
- **D.** $\oint_C \sec z\, dz \neq 0$

Q5.24.1 MCQ 2024 2M

Let z be a complex variable. If $f(z) = \frac{\sin(\pi z)}{z^2(z-2)}$ and C is the circle in the complex plane with $|z| = 3$ then $\oint_C f(z)\, dz$ is _____.

A. $\pi^2 j$ **B.** $j\pi\left(\frac{1}{2} - \pi\right)$ **C.** $j\pi\left(\frac{1}{2} + \pi\right)$ **D.** $-\pi^2 j$

Q5.23.1 MCQ 2023 1M

Let $w^4 = 16j$. Which of the following cannot be a value of w?

A. $2e^{\frac{j2\pi}{8}}$ **B.** $2e^{\frac{j\pi}{8}}$ **C.** $2e^{\frac{j5\pi}{8}}$ **D.** $2e^{\frac{j9\pi}{8}}$

Q5.23.2 MCQ 2023 1M

The value of the contour integral, $\oint_C \left(\frac{z+2}{z^2+2z+2}\right) dz$, where the contour C is $\left\{z : \left|z + 1 - \frac{3}{2}j\right| = 1\right\}$, taken in the counter clockwise direction, is

A. $-\pi(1 + j)$ **B.** $\pi(1 + j)$
C. $\pi(1 - j)$ **D.** $-\pi(1 - j)$

Q5.22.1 NAT 2022 1M

A simple closed path C in the complex plane is shown in the figure. If

$$\oint_C \frac{2^z}{z^2 - 1}\, dz = -i\pi A,$$

where $i = \sqrt{-1}$, then the value of A is _____ (rounded off to two decimal places).

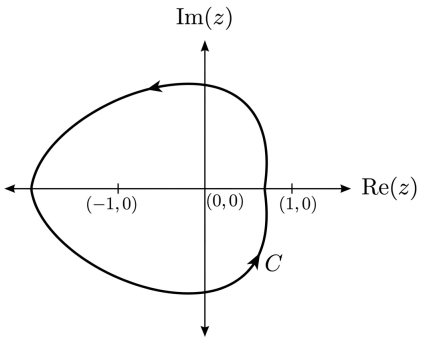


Fig. Q5.22.1

Q5.21.1 MCQ 2021 2M

Consider the integral

$$\oint_C \frac{\sin(x)}{x^2(x^2 + 4)}\, dx$$

where C is a counter-clockwise oriented circle defined as $|x - i| = 2$. The value of the integral is

A. $-\frac{\pi}{8}\sin(2i)$ **B.** $\frac{\pi}{8}\sin(2i)$
C. $-\frac{\pi}{4}\sin(2i)$ **D.** $\frac{\pi}{4}\sin(2i)$

Q5.19.1 MCQ 2019 1M

Which one of the following functions is analytic over the entire complex plane?

A. $\ln(z)$ **B.** $e^{1/z}$ **C.** $\frac{1}{1 - z}$ **D.** $\cos(z)$

Q5.19.2 NAT 2019 1M

The value of the contour integral

$$\frac{1}{2\pi j}\oint \left(z + \frac{1}{z}\right)^2 dz$$

evaluated over the unit circle $|z| = 1$ is _____.

Q5.18.1 NAT 2018 2M

The contour C given below is on the complex plane $z = x + jy$, where $j = \sqrt{-1}$. The value of the integral $\frac{1}{\pi j}\oint_C \frac{dz}{z^2 - 1}$ is _____.

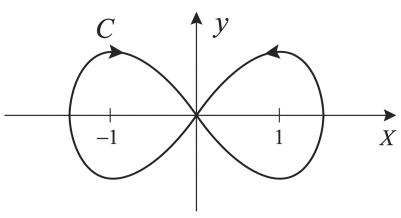


Fig. Q5.18.1

Q5.17.1 MCQ 2017 1M

The residues of a function

$$f(z) = \frac{1}{(z-4)(z+1)^3}$$

are

A. $\frac{-1}{27}$ and $\frac{-1}{125}$ **B.** $\frac{1}{125}$ and $\frac{-1}{125}$

C. $\frac{-1}{27}$ and $\frac{1}{5}$ **D.** $\frac{1}{125}$ and $\frac{-1}{5}$

Q5.17.2 MCQ 2017 1M

An integral I over a counterclockwise circle C is given by

$$I = \oint_C \frac{z^2-1}{z^2+1} e^z dz.$$

If C is defined as $|z| = 3$, then the value of I is

A. $-\pi i \sin(1)$ **B.** $-2\pi i \sin(1)$

C. $-3\pi i \sin(1)$ **D.** $-4\pi i \sin(1)$

Q5.16.1 NAT 2016 2M

In the following integral, the contour C encloses the points $2\pi j$ and $-2\pi j$

$$-\frac{1}{2\pi} \oint_C \frac{\sin z}{(z-2\pi j)^3} dz$$

The value of the integral is _____

Q5.16.2 NAT 2016 1M

Consider the complex valued function $f(z) = 2z^3 + b|z|^3$ where z is a complex variable. The value of b for which the function $f(z)$ is analytic is _____.

Q5.16.3 MCQ 2016 2M

The values of the integral $\frac{1}{2\pi j} \oint_c \frac{e^z}{z-2} dz$ along a closed contour c in anti-clockwise direction for
(i) The point $z_0 = 2$ inside the contour c, and
(ii) The point $z_0 = 2$ outside the contour c,

Respectively, are

A. (i) 2.72 (ii) 0 **B.** (i) 7.39 (ii) 0

C. (i) 0 (ii) 2.72 **D.** (i) 0 (ii) 7.39

Q5.15.1 MCQ 2015 1M

Let $z = x + iy$ be a complex variable. Consider that contour integration is performed along the unit circle in anticlockwise direction. Which one of the following statements is NOT TRUE?

A. The residue of $\frac{z}{z^2-1}$ at $z = 1$ is $1/2$

B. $\oint_C z^2 dz = 0$

C. $\frac{1}{2\pi i} \oint_C \frac{1}{z} dz = 1$

D. $\bar{z}$ (complex conjugate of z) is an analytical function

Q5.15.2 MCQ 2015 1M

The value of x for which all the eigen-values of the matrix given below are real is

$$\begin{bmatrix} 10 & 5+j & 4 \\ x & 20 & 2 \\ 4 & 2 & -10 \end{bmatrix}$$

A. $5 + j$ **B.** $5 - j$ **C.** $1 - 5j$ **D.** $1 + 5j$

Q5.15.3 NAT 2015 1M

Let $f(z) = \frac{az+b}{cz+d}$. If $f(z_1) = f(z_2)$ for all $z_1 \neq z_2$, $a = 2$, $b = 4$ and $c = 5$, then d should be equal to _____.

Q5.15.4 NAT 2015 2M

If C denotes the counterclockwise unit circle, the value of the contour integral

$$\frac{1}{2\pi j} \oint_C Re\{z\} dz$$

is _____.

Q5.15.5 MCQ 2015 1M

If C is a circle of radius r with centre z_0, in the complex z-plane and if n is a non-zero integer, then $\oint_C \frac{dz}{(z-z_0)^{n+1}}$ equals

A. $2\pi n j$ **B.** 0 **C.** $\frac{nj}{2\pi}$ **D.** $2\pi n$

Q5.14.1 MCQ 2014 2M

The real part of an analytic function $f(z)$ where $z = x + jy$ is given by $e^{-y}\cos(x)$. The imaginary part of $f(z)$ is

A. $e^{y}\cos(x)$ **B.** $e^{-y}\sin(x)$

C. $-e^{y}\sin(x)$ **D.** $-e^{-y}\sin(x)$

Q5.14.2 MCQ 2014 1M

For an all-pass system $H(z) = \dfrac{(z^{-1} - b)}{(1 - az^{-1})}$, where $|H(e^{-j\omega})| = 1$, for all ω. If $\mathrm{Re}(a) \neq 0$, $\mathrm{Im}(a) \neq 0$, then b equals

A. a **B.** a^{*} **C.** $1/a^{*}$ **D.** $1/a$

Q5.14.3 NAT 2014 2M

Let $H_1(z) = (1 - pz^{-1})^{-1}$, $H_2(z) = (1 - qz^{-1})^{-1}$, $H(z) = H_1(z) + rH_2(z)$. The quantities p, q, r are real numbers. Consider $p = \dfrac{1}{2}$, $q = -\dfrac{1}{4}$, $|r| < 1$. If the zero of $H(z)$ lies on the unit circle, then $r =$ _____

Q5.12.1 MCQ 2012 1M

Given $f(z) = \dfrac{1}{z+1} - \dfrac{2}{z+3}$. If C is a counterclockwise path in the z-plane such that $|z + 1| = 1$, the value of $\dfrac{1}{2\pi j}\oint_C f(z)\,dz$ is

A. -2 **B.** -1 **C.** 1 **D.** 2

Q5.11.1 MCQ 2011 1M

The value of the integral $\oint_C \dfrac{-3z + 4}{(z^2 + 4z + 5)}\,dz$ where C is the circle $|z| = 1$ is given by

A. 0 **B.** $1/10$ **C.** $4/5$ **D.** 1

Q5.10.1 MCQ 2010 2M

The residues of a complex function $X(z) = \dfrac{1 - 2z}{z(z-1)(z-2)}$ at its poles are

A. $\dfrac{1}{2}, -\dfrac{1}{2}$ and 1 **B.** $\dfrac{1}{2}, \dfrac{1}{2}$ and -1

C. $\dfrac{1}{2}$, 1 and $-\dfrac{3}{2}$ **D.** $\dfrac{1}{2}$, -1 and $\dfrac{3}{2}$

Q5.09.1 MCQ 2009 1M

If $f(z) = c_0 + c_1 z^{-1}$, then $\oint_{\substack{\text{unit}\\ \text{circle}}} \dfrac{1 + f(z)}{z}\,dz$ is given by

A. $2\pi c_1$ **B.** $2\pi(1 + c_0)$ **C.** $2\pi j c_1$ **D.** $2\pi j(1 + c_0)$

Q5.08.1 MCQ 2008 1M

The equation $\sin(z) = 10$ has

A. no real or complex solution

B. exactly two distinct complex solutions

C. a unique solution

D. an infinite number of complex solutions

Q5.08.2 MCQ 2008 2M

The residue of the function $f(z) = \dfrac{1}{(z+2)^2(z-2)^2}$ at $z = 2$ is

A. $-\dfrac{1}{32}$ **B.** $-\dfrac{1}{16}$ **C.** $\dfrac{1}{16}$ **D.** $\dfrac{1}{32}$

Q5.07.1 MCQ 2007 2M

If the semi-circular contour D of radius 2 is as shown in the figure, then the value of the integral $\oint_D \dfrac{1}{(s^2 - 1)}\,ds$ is

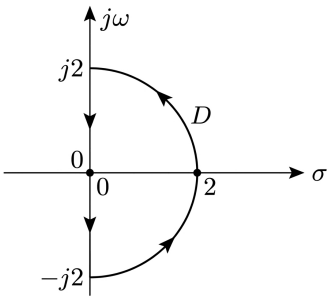


Fig. Q5.07.1

A. $j\pi$ **B.** $-j\pi$ **C.** $-\pi$ **D.** π

Q5.06.1 MCQ 2006 2M

For the function of a complex variable $W = \ln Z$ (where, $W = u + jv$ and $Z = x + jy$), then $u =$ constant lines get mapped in Z-plane as

A. Set of radial straight lines **B.** Set of concentric circles

C. Set of confocal hyperbolas **D.** Set of confocal ellipses

Q5.06.2 MCQ 2006 2M

The value of the constant integral $\oint_{|z-j|=2} \frac{1}{z^2+4} dz$ in positive sense is

A. $\frac{j\pi}{2}$ **B.** $\frac{-\pi}{2}$ **C.** $\frac{-j\pi}{2}$ **D.** $\frac{\pi}{2}$

Electrical Engineering

Q5.26.1 NAT 2026 2M

The magnitude of the contour integral

$$\oint_C \left(\frac{(z+1)^2}{(z-i)(z-2)} \right) dz$$

over the contour $C : |z-2-i| = 3/2$ is _____ (Round off to two decimal places) Note: z is a complex variable and $i = \sqrt{-1}$

Q5.25.1 NAT 2025 2M

Let C be a clockwise oriented closed curve in the complex plane defined by $|z| = 1$. Further, let $f(z) = jz$ be a complex function, where $j = \sqrt{-1}$. Then, $\oint_C f(z)dz =$ _____ (round off to the nearest integer).

Q5.24.1 MSQ 2024 1M

Which of the following complex functions is/are analytic on the complex plane?

- ☐ **A.** $f(z) = j\text{Re}(z)$
- ☐ **B.** $f(z) = \text{Im}(z)$
- ☐ **C.** $f(z) = e^{|z|}$
- ☐ **D.** $f(z) = z^2 - z$

Q5.24.2 NAT 2024 1M

Consider the complex function $f(z) = \cos z + e^{z^2}$. The coefficient of z^5 in the Taylor series expansion of $f(z)$ about the origin is _____ (rounded off to 1 decimal place).

Q5.21.1 MCQ 2021 1M

Let $p(z) = z^3 + (1+j)z^2 + (2+j)z + 3$, where z is a complex number. Which one of the following is true?

A. conjugate$\{p(z)\} = p(\text{conjugate}\{z\})$ for all z

B. The sum of the roots of $p(z) = 0$ is a real number

C. The complex roots of the equation $p(z) = 0$ come in conjugate pairs

D. All the roots cannot be real

Q5.21.2 MCQ 2021 2M

Let $(-1-j)$, $(3-j)$, $(3+j)$ and $(-1+j)$ be the vertices of a rectangle C in the complex plane. Assuming that C is traversed in counter-clockwise direction, the value of the contour integral $\oint_C \frac{dz}{z^2(z-4)}$ is

A. $j\pi/2$ **B.** 0 **C.** $-j\pi/8$ **D.** $j\pi/16$

Q5.20.1 MCQ 2020 1M

The value of the following complex integral, with C representing the unit circle centered at origin in the counter-clockwise sense, is:

$$\int_C \frac{z^2+1}{z^2-2z} dz$$

A. $8\pi i$ **B.** $-8\pi i$ **C.** $-\pi i$ **D.** πi

Q5.19.1 MCQ 2019 1M

Which one of the following functions is analytic in the region $|z| \leq 1$?

A. $\frac{z^2-1}{z}$ **B.** $\frac{z^2-1}{z+2}$ **C.** $\frac{z^2-1}{z-0.5}$ **D.** $\frac{z^2-1}{z+j0.5}$

Q5.19.2 MCQ 2019 2M

The closed loop line integral

$$\oint_{|z|=5} \frac{z^3+z^2+8}{z+2} dz$$

evaluated counter-clockwise, is

A. $+8j\pi$ **B.** $-8j\pi$ **C.** $-4j\pi$ **D.** $+4j\pi$

Q5.18.1 MCQ 2018 1M

The value of the integral $\oint_C \frac{z+1}{z^2-4} dz$ in counter clockwise direction around a circle C of radius 1 with center at the point $z = -2$ is

A. $\frac{\pi i}{2}$ **B.** $2\pi i$ **C.** $-\frac{\pi i}{2}$ **D.** $-2\pi i$

Q5.18.2 MCQ 2018 2M

If C is a circle $|z| = 4$ and $f(z) = \frac{z^2}{(z^2-3z+2)^2}$, then $\oint_C f(z)dz$ is

A. 1 **B.** 0 **C.** -1 **D.** -2

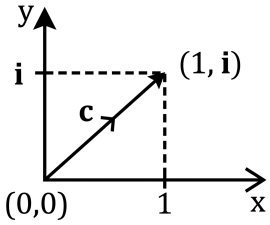
Q5.17.1 MCQ 2017 1M

For a complex number z , $\lim_{z\to i} \frac{z^2+1}{z^3+2z-i(z^2+2)}$ is

A. $-2i$ **B.** $-i$ **C.** i **D.** $2i$

Q5.17.2 MCQ 2017 2M

Consider the line integral $I = \int_c (x^2 + iy^2)\, dz$, where $z = x + iy$. The line c is shown in the figure below.



The value of I is

A. $\frac{1}{2}i$ **B.** $\frac{2}{3}i$ **C.** $\frac{3}{4}i$ **D.** $\frac{4}{5}i$

Q5.16.1 MCQ 2016 1M

The value of the integral

$$\oint_C \frac{2z+5}{\left(z-\frac{1}{2}\right)\left(z^2-4z+5\right)} dz$$

over the contour $|z| = 1$, taken in the anti-clockwise direction, would be

A. $\frac{24\pi i}{13}$ **B.** $\frac{48\pi i}{13}$ **C.** $\frac{24}{13}$ **D.** $\frac{12}{13}$

Q5.16.2 MCQ 2016 1M

Consider the function $f(z) = z + z^*$ where z is a complex variable and z^* denotes it complex conjugate. Which one of the following is TRUE?

A. $f(z)$ is both continuous and analytic

B. $f(z)$ is continuous but not analytic

C. $f(z)$ is not continuous but is analytic

D. $f(z)$ is neither continuous nor analytic

Q5.15.1 MCQ 2015 1M

Given $f(z) = g(z) + h(z)$, where f, g, h are complex valued functions of a complex variable z. Which one of the following statements is TRUE?

A. If $f(z)$ is differentiable at z_0, then $g(z)$ and $h(z)$ are also differentiable at z_0.

B. If $g(z)$ and $h(z)$ are differentiable at z_0, then $f(z)$ is also differentiable at z_0.

C. If $f(z)$ is continuous at z_0, then it is differentiable at z_0.

D. If $f(z)$ is differentiable at z_0, then so are its real and imaginary parts.

Q5.14.1 MCQ 2014 1M

Let S be the set of points in the complex plane corresponding to the unit circle. (That is, $S = \{z : |z| = 1\}$). Consider the function $f(z) = zz^*$ where z^* denotes the complex conjugate of z. The $f(z)$ maps S to which one of the following in the complex plane

A. unit circle

B. horizontal axis line segment from origin to $(1, 0)$

C. the point $(1, 0)$

D. the entire horizontal axis

Q5.14.2 MCQ 2014 1M

All the values of the multi-valued complex function 1^i, where $i = \sqrt{-1}$, are

A. purely imaginary. **B.** real and non-negative.

C. on the unit circle. **D.** equal in real and imaginary parts.

Q5.14.3 MCQ 2014 2M

Integration of the complex function $f(z) = \frac{z^2}{z^2-1}$, in the counterclockwise direction, around $|z - 1| = 1$, is

A. $-\pi i$ **B.** 0 **C.** πi **D.** $2\pi i$

Q5.13.1 MCQ 2013 1M

Square roots of $-i$, where $i = \sqrt{-1}$, are

A. $i, -i$

B. $\cos(-\frac{\pi}{4}) + i\sin(-\frac{\pi}{4}), \cos(\frac{3\pi}{4}) + i\sin(\frac{3\pi}{4})$

C. $\cos(\frac{\pi}{4}) + i\sin(\frac{3\pi}{4}), \cos(\frac{3\pi}{4}) + i\sin(\frac{\pi}{4})$

D. $\cos(\frac{3\pi}{4}) + i\sin(-\frac{3\pi}{4}), \cos(-\frac{3\pi}{4}) + i\sin(\frac{3\pi}{4})$

Q5.13.2 MCQ 2013 2M

$\oint \frac{z^2-4}{z^2+4}\,dz$ evaluated anticlockwise around the circle $|z-i| = 2$, where $i = \sqrt{-1}$, is

A. -4π **B.** 0 **C.** $2+\pi$ **D.** $2+2i$

Q5.12.1 MCQ 2012 1M

If $x = \sqrt{-1}$, then the value of x^x is

A. $e^{-\pi/2}$ **B.** $e^{\pi/2}$ **C.** x **D.** 1

Q5.11.1 MCQ 2011 1M

Roots of the algebraic equation $x^3 + x^2 + x + 1 = 0$ are

A. $(+1, +j, -j)$ **B.** $(+1, -1, +1)$

C. $(0, 0, 0)$ **D.** $(-1, +j, -j)$

Q5.11.2 MCQ 2011 1M

A point z has been plotted in the complex plane, as shown in figure below. The plot of the complex number $y = \frac{1}{z}$ is

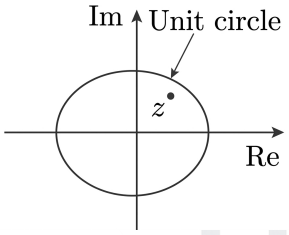
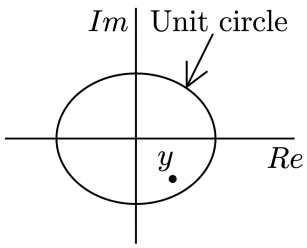
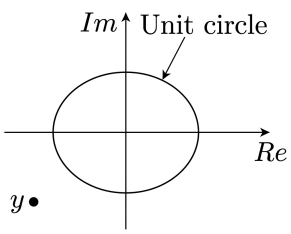
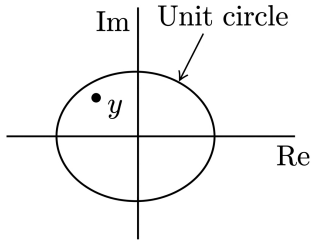
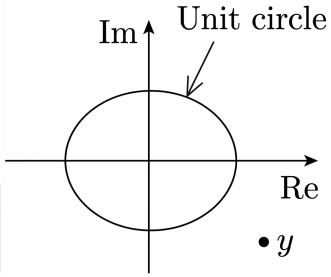


Fig. Q5.11.2

A.


B.


C.


D.


Q5.08.1 MCQ 2008 2M

Given $X(z) = \frac{z}{(z-a)^2}$ with $|z| > a$, the residue of $X(z)z^{n-1}$ at $z = a$ for $n \geq 0$ will be

A. a^{n-1} **B.** a^n **C.** na^n **D.** na^{n-1}

Q5.07.1 MCQ 2007 2M

The value of $\oint_C \frac{dz}{(1+z^2)}$ where C is the contour $|z - i/2| = 1$ is

A. $2\pi i$ **B.** π **C.** $\tan^{-1} z$ **D.** $\pi i \tan^{-1} z$

Q5.97.1 MCQ 1997 1M

e^z is a periodic with a period of

A. 2π **B.** $2\pi i$ **C.** π **D.** $i\pi$

Instrumentation Engineering

Q5.26.1 MCQ 2026 1M

Consider a function $f(z) = z^2 + z + 1$ where $z \in \mathbb{C}$ is a complex variable. A simple closed contour γ in z-plane encloses the point $z = 1 + 0j$. The value of integral $\oint_\gamma \frac{f(z)}{z-1}\,dz =$ _____.

A. $6\pi j$ **B.** $3\pi j$ **C.** $12\pi j$ **D.** πj

Q5.25.1 NAT 2025 1M

Consider the function $f(z) = \frac{2z+1}{z^2-z}$, where z is a complex variable. The sum of the residues at singular points of $f(z)$ is _____ (in integer).

Q5.25.2 MSQ 2025 2M

Choose the correct statement(s) from the following options, regarding Cauchy's theorem on complex integration $\oint_C f(z)\,dz$ where C is a simple closed path in a simply connected domain D.

- ☐ **A.** Cauchy's theorem cannot be directly applied to conclude that $\oint_C f(z)\,dz = 0$ when $f(z) = \frac{1}{z^2}$, and C is the unit circle
- ☐ **B.** If $f(z)$ is analytic in D, then it can be concluded that $\oint_C f(z)\,dz = 0$ for any simple closed path C in D
- ☐ **C.** The function $f(z)$ must be analytic in D to conclude $\oint_C f(z)\,dz = 0$ for any simple closed path C in D
- ☐ **D.** $\oint_C f(z)\,dz \neq 0$ when $f(z) = \frac{1}{z^2}$, since the function is not analytic at $z = 0$

Q5.24.1 MCQ 2024 1M

Let $z = x + iy$ be a complex variable and $\bar{z}$ be its complex conjugate. The equation $\bar{z}^2 + z^2 = 2$ represents a

A. parabola **B.** hyperbola **C.** ellipse **D.** circle

Q5.24.2 MSQ 2024 2M

The complex functions $f(z) = u(x,y) + i\,v(x,y)$ and $\overline{f(z)} = u(x,y) - i\,v(x,y)$ are both analytic in a given domain. Choose the correct option(s) from the following.

A. $\frac{\partial u}{\partial x} = \frac{\partial v}{\partial y} = 0$ **B.** $\frac{\partial u}{\partial y} = -\frac{\partial v}{\partial x} \neq 0$

C. $\frac{df(z)}{dz} = 0$ **D.** $\frac{df(z)}{dz} \neq 0$

Q5.23.1 MCQ 2023 2M

Let $f(z) = j\frac{1-z}{1+z}$, where z denotes a complex number and j denotes $\sqrt{-1}$. The inverse function $f^{-1}(z)$ maps the real axis to the _____.

A. unit circle with centre at the origin

B. unit circle with centre not at the origin

C. imaginary axis

D. real axis

Q5.22.1 MSQ 2022 2M

For the complex number $z = \frac{a+jb}{a-jb}$, where $a > 0$ and $b > 0$. Which of the following statement(s) is/are true?

- ☐ **A.** The phase is $2\tan^{-1}\frac{b}{a}$
- ☐ **B.** The phase is $\tan^{-1}\frac{2b}{a}$
- ☐ **C.** The magnitude is 1
- ☐ **D.** The magnitude is $\sqrt{\frac{a^2+b^2}{a^2-b^2}}$

Q5.22.2 NAT 2022 2M

Consider the function $f(z) = \frac{1}{(z+1)(z+2)(z+3)}$. The residue of $f(z)$ at $z = -1$, is _____.

Q5.21.1 NAT 2021 1M

Let $f(z) = \frac{1}{z^2+6z+9}$ defined in the complex plane. The integral $\oint_c f(z)\,dz$ over the contour of a circle c with center at the origin and unit radius is _____.

Q5.21.2 MCQ 2021 2M

$f(z) = (z-1)^{-1} - 1 + (z-1) - (z-1)^2 + \cdots$ is the series expansion of

A. $\frac{-1}{z(z-1)}$ for $|z-1| < 1$

B. $\frac{1}{z(z-1)}$ for $|z-1| < 1$

C. $\frac{1}{(z-1)^2}$ for $|z-1| < 1$

D. $\frac{-1}{(z-1)}$ for $|z-1| < 1$

Q5.20.1 MCQ 2020 1M

Let $f(z) = \frac{1}{z+a}, a > 0$. The value of the integral $\oint f(z)dz$ over a circle C with center $(-a, 0)$ and radius $R > 0$ evaluated in the anti-clockwise direction is _____

A. 0 **B.** $2\pi i$ **C.** $-2\pi i$ **D.** $4\pi i$

Q5.19.1 MCQ 2019 2M

A complex function $f(z) = u(x, y) + i\,v(x, y)$ and its complex conjugate $f^*(z) = u(x, y) - i\,v(x, y)$ are both analytic in the entire complex plane, where $z = x + i\,y$ and $i = \sqrt{-1}$. The function f is then given by

A. $f(z) = x + i\,y$ **B.** $f(z) = x^2 - y^2 + i\,2xy$

C. $f(z) =$ constant **D.** $f(z) = x^2 + y^2$

Q5.18.1 MCQ 2018 1M

Let $f_1(z) = z^2$ and $f_2(z) = \bar{z}$ be two complex variable functions. Here $\bar{z}$ is the complex conjugate of z. Choose the correct answer.

A. Both $f_1(z)$ and $f_2(z)$ are analytic

B. Only $f_1(z)$ is analytic

C. Only $f_2(z)$ is analytic

D. Both $f_1(z)$ and $f_2(z)$ are not analytic

Q5.17.1 MCQ 2017 1M

Let $z = x + jy$ where $j = \sqrt{-1}$. Then $\overline{\cos z} =$

A. $\cos z$ **B.** $\cos \bar{z}$ **C.** $\sin z$ **D.** $\sin \bar{z}$

Q5.16.1 NAT 2016 2M

The value of the integral $\frac{1}{2\pi j} \oint_C \frac{z^2 + 1}{z^2 - 1}\,dz$ where z is a complex number and C is a unit circle with center at $1 + 0j$ in the complex plane is _____.

Q5.15.1 MCQ 2015 1M

The value of $\oint \frac{1}{z^2}\,dz$, where the contour is the unit circle traversed clockwise, is

A. $-2\pi i$ **B.** 0 **C.** $2\pi i$ **D.** $4\pi i$

Q5.13.1 MCQ 2013 1M

The complex function $\tanh(s)$ is analytic over a region of the imaginary axis of the complex s-plane if the following is **TRUE** everywhere in the region for all integers n

A. $Re(s) = 0$ **B.** $Im(s) \neq n\pi$

C. $Im(s) \neq \frac{n\pi}{3}$ **D.** $Im(s) \neq \frac{(2n+1)\pi}{2}$

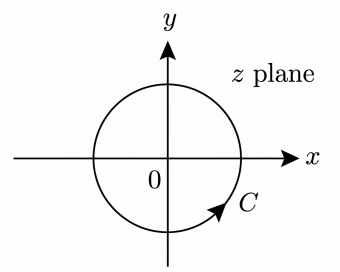
Q5.11.1 MCQ 2011 1M

The contour integral $\oint_C e^{1/z}\,dz$ with C as the counter-clockwise unit circle in the z-plane is equal to

A. 0 **B.** 2π **C.** $2\pi\sqrt{-1}$ **D.** ∞

Q5.10.1 MCQ 2010 1M

The contour C in the adjoining figure is described by $x^2 + y^2 = 16$. The value of $\oint_C \frac{z^2 + 8}{0.5z - 1.5j}\,dz$ is _____.(Note: $j = \sqrt{-1}$)



A. $-2\pi j$ **B.** $2\pi j$ **C.** $4\pi j$ **D.** $-4\pi j$

Q5.09.1 MCQ 2009 1M

If $z = x + jy$, where x and y are real, the value of $\left|e^{j\,z}\right|$ is

A. 1 **B.** $e^{\sqrt{x^2+y^2}}$ **C.** e^{y} **D.** e^{-y}

Q5.09.2 MCQ 2009 1M

The value of $\oint \frac{\sin z}{z}\,dz$, where the contour of integration is a simple closed curve around the origin, is

A. 0 **B.** $2\pi j$ **C.** ∞ **D.** $1/(2\pi j)$

Q5.09.3 MCQ 2009 2M

One of the roots of the equation $x^3 = j$, where j is the positive square root of -1, is

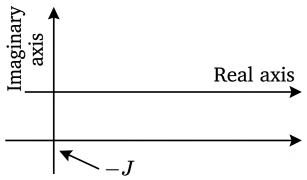
A. j **B.** $\frac{\sqrt{3}}{2} + j\frac{1}{2}$

C. $\frac{\sqrt{3}}{2} - j\frac{1}{2}$ **D.** $-\frac{\sqrt{3}}{2} - j\frac{1}{2}$

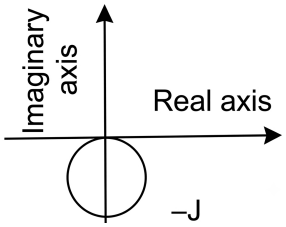
Q5.08.1 MCQ 2008 2M

A complex variable $z = x + \mathrm{j}\,0.1$ has its real part x varying in the range $-\infty$ to $+\infty$. Which one of the following is the locus (shown in thick lines) of $\frac{1}{z}$ in the complex plane?

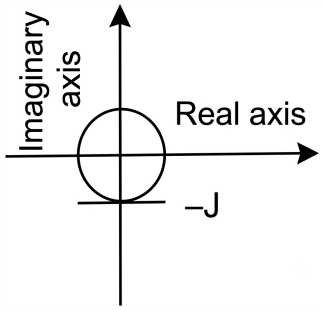
A.



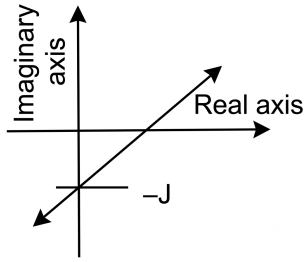
B.



C.



D.



Q5.07.1 MCQ 2007 2M

For the function $\frac{\sin z}{z^3}$ of a complex variable z, the point $z = 0$ is

A. a pole of order 3

B. a pole of order 2

C. a pole of order 1

D. not a singularity

Q5.05.1 MCQ 2005 2M

Consider the circle $|Z - 5 - 5i| = 2$ in the complex plane (x, y) with $Z = x + iy$. The minimum distance from the origin to the circle is

A. $5\sqrt{2} - 2$ **B.** $\sqrt{54}$ **C.** $\sqrt{34}$ **D.** $5\sqrt{2}$

Q5.05.2 MCQ 2005 2M

Let $Z^3 = \overline{Z}$, where Z is a complex number not equal to zero. Then Z is a solution of

A. $Z^2 = 1$ **B.** $Z^3 = 3$

C. $Z^4 = 1$ **D.** $Z^9 = 1$

Q5.02.1 MCQ 2002 1M

The bilinear transformation $w = \frac{z-1}{z+1}$

A. maps the inside of the unit circle in the z-plane to the left half of the w-plane

B. maps the outside the unit circle in the z-plane to the left half of the w-plane

C. maps the inside of the unit circle in the z-plane to right half of the w-plane

D. maps the outside of the unit circle in the z-plane to the right half of the w-plane

Q5.97.1 MCQ 1997 1M

The complex number $z = x + jy$ which satisfy the equation $|z + 1| = 1$ lie on

A. a circle with $(1, 0)$ as the centre and radius 1

B. a circle with $(-1, 0)$ as the centre and radius 1

C. y-axis

D. x-axis

Q5.94.1 MCQ 1994 1M

The real part of the complex number $z = x + iy$ is given by

A. $\text{Re}(z) = z - z^*$ **B.** $\text{Re}(z) = \frac{z - z^*}{2}$

C. $\text{Re}(z) = \frac{z + z^*}{2}$ **D.** $\text{Re}(z) = z + z^*$

Q5.94.2 MCQ 1994 1M

$\cos\phi$ can be represented as

A. $\frac{e^{i\phi} - e^{-i\phi}}{2}$ **B.** $\frac{e^{i\phi} - e^{-i\phi}}{2i}$

C. $\frac{e^{i\phi} + e^{-i\phi}}{i}$ **D.** $\frac{e^{i\phi} + e^{-i\phi}}{2}$

Mechanical Engineering

Q5.26.1 MCQ 2026 2M

If $w = \log_e z = \log_e(x + iy)$, where $i = \sqrt{-1}$, then which one of the following statements is correct?

A. w is analytic everywhere except at $z = 0$

B. w is non-analytic everywhere

C. The conjugate functions of w are $\log_e(x^2 + y^2)$ and $\log_e(x^2 - y^2)$

D. The conjugate functions of w are $\tan^{-1}(x/y)$ and $\tan^{-1}(y/x)$

Q5.25.1 NAT 2025 2M

If C is the unit circle in the complex plane with its center at the origin, then the value of n in the equation given below is _____ (rounded off to 1 decimal place).

$$\oint_C \frac{z^3}{(z^2+4)(z^2-4)}dz = 2\pi i n$$

Q5.24.1 MCQ 2024 1M

Let $f(z)$ be an analytic function, where $z = x + iy$. If the real part of $f(z)$ is $\cosh x \cos y$, and the imaginary part of $f(z)$ is zero for $y = 0$, then $f(z)$ is

A. $\cosh x \exp(-iy)$ **B.** $\cosh z \exp z$

C. $\cosh z \cos y$ **D.** $\cosh z$

Q5.23.1 NAT 2023 1M

The value of k that makes the complex-valued function

$$f(z) = e^{-kx}(\cos 2y - i \sin 2y)$$

analytic, where $z = x + iy$, is _____. (Answer in integer)

Q5.22.1 MCQ 2022 2M

The value of the integral

$$\oint \left(\frac{6z}{2z^4 - 3z^3 + 7z^2 - 3z + 5}\right) dz$$

evaluated over a counter-clockwise circular contour in the complex plane enclosing only the pole $z = i$, where i is the imaginary unit, is

A. $(-1 + i)\pi$ **B.** $(1 + i)\pi$

C. $2(1 - i)\pi$ **D.** $(2 + i)\pi$

Q5.22.2 NAT 2022 2M

Given $z = x+iy, i = \sqrt{-1}$. C is a circle of radius 2 with the centre at the origin. If the contour C is traversed anticlockwise, then the value of the integral $\frac{1}{2\pi}\int_C \frac{1}{(z-i)(z+4i)}dz$ is _____ (round off to one decimal place).

Q5.21.1 MCQ 2021 2M

Let C represent the unit circle centered at origin in the complex plane, and complex variable, $z = x+iy$. The value of the contour integral $\oint_C \frac{\cosh 3z}{2z}dz$ (where integration is taken counter clockwise) is

A. 0 **B.** 2 **C.** πi **D.** $2\pi i$

Q5.21.2 MCQ 2021 1M

Value of $(1 + i)^8$, where $i = \sqrt{-1}$, is equal to

A. 4 **B.** 16 **C.** $4i$ **D.** $16i$

Q5.20.1 MCQ 2020 1M

Which of the following function $f(z)$, of the complex variable z, is **NOT** analytic at all the points of the complex plane?

A. $f(z) = z^2$ **B.** $f(z) = e^z$

C. $f(z) = \sin z$ **D.** $f(z) = \log z$

Q5.20.2 NAT 2020 2M

An analytic function of a complex variable $z = x + iy$ $(i = \sqrt{-1})$ is defined as

$$f(z) = x^2 - y^2 + i\psi(x, y),$$

where $\psi(x, y)$ is a real function. The value of the imaginary part of $f(z)$ at $z = (1 + i)$ is _____ (round off to 2 decimal places).

Q5.20.3 MCQ 2020 2M

The function $f(z)$ of complex variable $z = x + iy$, where $i = \sqrt{-1}$, is given as $f(z) = (x^3 - 3xy^2) + iv(x, y)$. For this function to be analytic, $v(x, y)$ should be

A. $(3xy^2 - y^3)$ + constant

B. $(3x^2y^2 - y^3)$ + constant

C. $(x^3 - 3x^2y)$ + constant

D. $(3x^2y - y^3)$ + constant

Q5.19.1 MCQ 2019 1M

An analytic function $f(z)$ of complex variable $z = x + iy$ may be written as $f(z) = u(x, y) + iv(x, y)$. Then, $u(x, y)$ and $v(x, y)$ must satisfy

A. $\frac{\partial u}{\partial x} = \frac{\partial v}{\partial y}$ and $\frac{\partial u}{\partial y} = \frac{\partial v}{\partial x}$

B. $\frac{\partial u}{\partial x} = \frac{\partial v}{\partial y}$ and $\frac{\partial u}{\partial y} = -\frac{\partial v}{\partial x}$

C. $\frac{\partial u}{\partial x} = -\frac{\partial v}{\partial y}$ and $\frac{\partial u}{\partial y} = \frac{\partial v}{\partial x}$

D. $\frac{\partial u}{\partial x} = -\frac{\partial v}{\partial y}$ and $\frac{\partial u}{\partial y} = -\frac{\partial v}{\partial x}$

Q5.18.1 MCQ 2018 2M

Let z be a complex variable. For a counter-clockwise integration around a unit circle C, centred at origin,

$$\oint_C \frac{1}{5z-4} dz = A\pi i,$$

the value of A is

A. 2/5 **B.** 1/2 **C.** 2 **D.** 4/5

Q5.17.1 MCQ 2017 2M

If $f(z) = (x^2 + ay^2) + ibxy$ is a complex analytic function of $z = x + iy$, where $i = \sqrt{-1}$, then

A. $a = -1, b = -1$ **B.** $a = -1, b = 2$

C. $a = 1, b = 2$ **D.** $a = 2, b = 2$

Q5.16.1 MCQ 2016 1M

$f(z) = u(x, y) + iv(x, y)$ is an analytic function of complex variable $z = x + iy$ where $i = \sqrt{-1}$. If $u(x, y) = 2xy$, then $v(x, y)$ may be expressed as

A. $-x^2 + y^2$ + constant **B.** $x^2 - y^2$ + constant

C. $x^2 + y^2$ + constant **D.** $-(x^2 + y^2)$ + constant

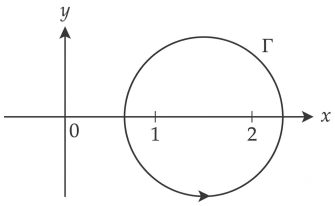
Q5.16.2 NAT 2016 1M

A function f of the complex variable $z = x + iy$, is given as $f(x, y) = u(x, y) + iv(x, y)$, where $u(x, y) = 2kxy$ and $v(x, y) = x^2 - y^2$. The value of k, for which the function is analytic, is _____.

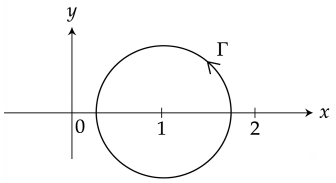
Q5.16.3 MCQ 2016 2M

The value of $\oint_\Gamma \frac{3z-5}{(z-1)(z-2)} dz$ along a closed path Γ is equal to $(4\pi i)$, where $z = x + iy$ and $i = \sqrt{-1}$. The correct path Γ is

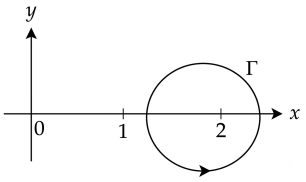
A.



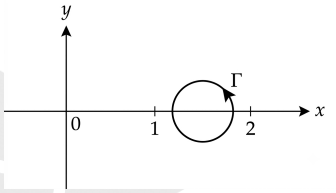
B.



C.



D.



Q5.15.1 MCQ 2015 1M

Given two complex numbers $z_1 = 5 + (5\sqrt{3})i$ and $z_2 = \frac{2}{\sqrt{3}} + 2i$, the argument of $\frac{z_1}{z_2}$ in degrees is

A. 0 **B.** 30 **C.** 60 **D.** 90

Q5.14.1 MCQ 2014 1M

The argument of the complex number $\frac{1+i}{1-i}$, where $i = \sqrt{-1}$, is

A. $-\pi$ **B.** $-\frac{\pi}{2}$ **C.** $\frac{\pi}{2}$ **D.** π

Q5.14.2 MCQ 2014 2M

An analytic function of a complex variable $z = x + iy$ is expressed as $f(z) = u(x, y) + iv(x, y)$, where $i = \sqrt{-1}$. If $u(x, y) = x^2 - y^2$, then expression for $v(x, y)$ in terms of x, y and a general constant c would be

A. $xy + c$ **B.** $\frac{x^2 + y^2}{2} + c$

C. $2xy + c$ **D.** $\frac{(x-y)^2}{2} + c$

Q5.11.1 MCQ 2011 1M

The product of two complex numbers $1 + i$ and $2 - 5i$ is

A. $7 - 3i$ **B.** $3 - 4i$

C. $-3 - 4i$ **D.** $7 + 3i$

Q5.10.1 MCQ 2010 1M

The modulus of the complex number $\left(\frac{3+4i}{1-2i}\right)$ is

A. 5 **B.** $\sqrt{5}$ **C.** $1/\sqrt{5}$ **D.** $1/5$

Q5.09.1 MCQ 2009 2M

An analytic function of a complex variable $z = x + iy$ is expressed as $f(z) = u(x, y) + iv(x, y)$ where $i = \sqrt{-1}$. If $u = xy$, the expression for v should be

A. $\frac{(x+y)^2}{2} + k$

B. $\frac{x^2 - y^2}{2} + k$

C. $\frac{y^2 - x^2}{2} + k$

D. $\frac{(x-y)^2}{2} + k$

Q5.08.1 MCQ 2008 2M

The integral $\oint f(z)\,dz$ evaluated around the unit circle on the complex plane for $f(z) = \frac{\cos z}{z}$ is

A. $2\pi i$ **B.** $4\pi i$ **C.** $-2\pi i$ **D.** 0

Q5.07.1 MCQ 2007 1M

If $\phi(x, y)$ and $\psi(x, y)$ are functions with continuous second derivatives, then $\phi(x, y) + i\psi(x, y)$ can be expressed as an analytic function of $x + iy$ $(i = \sqrt{-1})$, when

A. $\frac{\partial\phi}{\partial x} = -\frac{\partial\psi}{\partial x}; \frac{\partial\phi}{\partial y} = \frac{\partial\psi}{\partial y}$

B. $\frac{\partial\phi}{\partial y} = -\frac{\partial\psi}{\partial x}; \frac{\partial\phi}{\partial x} = \frac{\partial\psi}{\partial y}$

C. $\frac{\partial^2\phi}{\partial x^2} + \frac{\partial^2\phi}{\partial y^2} = \frac{\partial^2\psi}{\partial x^2} + \frac{\partial^2\psi}{\partial y^2} = 1$

D. $\frac{\partial\phi}{\partial x} + \frac{\partial\phi}{\partial y} = \frac{\partial\psi}{\partial x} + \frac{\partial\psi}{\partial y} = 0$

Q5.06.1 MCQ 2006 2M

Assuming $i = \sqrt{-1}$ and t is a real number $\int_0^{\pi/3} e^{it}\,dt$ is

A. $\frac{\sqrt{3}}{2} + i\frac{1}{2}$ **B.** $\frac{\sqrt{3}}{2} - i\frac{1}{2}$

C. $\frac{1}{2} + i\left(\frac{\sqrt{3}}{2}\right)$ **D.** $\frac{1}{2} + i\left(1 + \frac{\sqrt{3}}{2}\right)$

Civil Engineering

Q5.11.1 MCQ 2011 1M

For an analytic function, $f(x+iy) = u(x, y) + iv(x, y)$, u is given by $u = 3x^2 - 3y^2$. The expression for v, considering K to be constant is

A. $3y^2 - 3x^2 + K$ **B.** $6x - 6y + K$

C. $6y - 6x + K$ **D.** $6xy + K$

Q5.10.1 MCQ 2010 1M

The value of the integral $\int_{-\infty}^{\infty} \frac{dx}{1+x^2}$ is

A. $-\pi$ **B.** $-\pi/2$ **C.** $\pi/2$ **D.** π

Q5.09.1 MCQ 2009 1M

The analytic function has singularities at, where $f(z) = \frac{z-1}{z^2+1}$

A. 1 & -1 **B.** 1 and i

C. 1 and $-$ i **D.** i and $-$ i

Q5.09.2 MCQ 2009 2M

The value of the integral $\oint_C \frac{\cos(2\pi z)}{(2z-1)(z-3)}\,dz$ where c is a closed curve given by $|z| = 1$ is

A. $-\pi i$ **B.** $\frac{\pi i}{5}$ **C.** $\frac{2\pi i}{5}$ **D.** πi

Q5.07.1 MCQ 2007 2M

Potential function ϕ is given as $\phi = x^2 - y^2$, what will be the stream function ψ with the condition $\psi = 0$ at $x = y = 0$?

A. $2xy$ **B.** $x^2 + y^2$

C. $x^2 - y^2$ **D.** $2x^2y^2$

Q5.06.1 MCQ 2006 1M

Using Cauchy's integral theorem, the value of the integral (integration being taken in counter clockwise direction) $\oint_C \frac{z^3 - 6}{3z - i}\,dz$ is, where C is a simple closed contour enclosing $z = i/3$ once:

A. $\frac{2\pi}{81} - 4\pi i$ **B.** $\frac{\pi}{8} - 6\pi i$ **C.** $\frac{4\pi}{81} - 6\pi i$ **D.** 1

Q5.05.1 MCQ 2005 1M

Which one of the following is Not true for complex numbers Z_1 and Z_2 ?

A. $\dfrac{Z_1}{Z_2} = \dfrac{Z_1\overline{Z_2}}{|Z_2|^2}$

B. $|Z_1 + Z_2| \le |Z_1| + |Z_2|$

C. $|Z_1 - Z_2| \le |Z_1| - |Z_2|$

D. $|Z_1 + Z_2|^2 + |Z_1 - Z_2|^2 = 2|Z_1|^2 + 2|Z_2|^2$

Q5.05.2 MCQ 2005 2M

Consider likely applicability of cauchy's Integral theorem to evaluate the following integral counter clockwise around the unit circle C.

$$I = \oint_C \sec z \, dz$$

z being a complex variable. The value of I will be

A. $I = 0$: singularities set = ϕ

B. $I = 0$: singularities set = $\left\{\pm\dfrac{2n+1}{2}\pi / n = 0, 1, 2, \ldots\right\}$

C. $I = \pi/2$: singularities set = $\{\pm n\pi : n = 0, 1, 2, \ldots\}$

D. None of above

Chemical Engineering

Q5.26.1 MCQ 2026 2M

Consider the following complex numbers

$$z_1 = r_1(\cos\theta_1 + i\sin\theta_1)$$

$$z_2 = r_2(\cos\theta_2 + i\sin\theta_2)$$

where r_1, r_2 are real numbers, $0 \le \theta_1 \le \dfrac{\pi}{2}$, $0 \le \theta_2 \le \dfrac{\pi}{2}$, and $i = \sqrt{-1}$

If $|z_1 + z_2| = |z_1| + |z_2|$, which one of the following conditions is necessarily CORRECT?

A. $\theta_1 = 0, \theta_2 = \dfrac{\pi}{2}$ **B.** $\theta_1 = \dfrac{\pi}{2}, \theta_2 = 0$

C. $r_1 = r_2$ **D.** $\theta_1 = \theta_2$

Q5.25.1 MCQ 2025 1M

Consider two complex numbers $z_1 = 1 - i$ and $z_2 = i$. The argument of $z_1 z_2$ is

A. 0 **B.** $\dfrac{\pi}{4}$ **C.** $\dfrac{\pi}{2}$ **D.** π

Q5.24.1 MCQ 2024 1M

If $z_1 = -1 + i$ and $z_2 = 2i$, where $i = \sqrt{-1}$, then $\text{Arg}(z_1/z_2)$ is

A. $\dfrac{3\pi}{4}$ **B.** $\dfrac{\pi}{4}$ **C.** $\dfrac{\pi}{2}$ **D.** $\dfrac{\pi}{3}$

Q5.23.1 MCQ 2023 1M

Given that

$$F = \frac{|z_1 + z_2|}{|z_1| + |z_2|},$$

where $z_1 = 2 + 3i$ and $z_2 = -2 + 3i$ with $i = \sqrt{-1}$, which one of the following options is CORRECT?

A. $F < 0$ **B.** $F < 1$ **C.** $F > 1$ **D.** $F = 1$

Q5.20.1 MCQ 2020 1M

Consider the hyperbolic functions in **Group – 1** and their definitions in **Group – 2**.

	Group – 1		**Group – 2**
P	$\tanh x$	**I**	$\dfrac{e^x + e^{-x}}{e^x - e^{-x}}$
Q	$\coth x$	**II**	$\dfrac{2}{e^x + e^{-x}}$
R	$\text{sech}\, x$	**III**	$\dfrac{2}{e^x - e^{-x}}$
S	$\text{cosech}\, x$	**IV**	$\dfrac{e^x - e^{-x}}{e^x + e^{-x}}$

The correct combination is

A. P – IV, Q – I, R – III, S – II

B. P – II, Q – III, R – I, S – IV

C. P – IV, Q – I, R – II, S – III

D. P – I, Q – II, R – IV, S – III

Q5.19.1 MCQ 2019 2M

The value of the complex number $i^{-1/2}$ (where $i = \sqrt{-1}$) is

A. $\frac{1}{\sqrt{2}}(1-i)$ **B.** $-\frac{1}{\sqrt{2}}i$ **C.** $\frac{1}{\sqrt{2}}i$ **D.** $\frac{1}{\sqrt{2}}(1+i)$

Q5.17.1 NAT 2017 1M

The real part of $6e^{i\pi/3}$ is _____.

Q5.16.1 MCQ 2016 1M

What are the modulus (r) and argument (θ) of the complex number $3 + 4i$?

A. $r = \sqrt{7}, \quad \theta = \tan^{-1}\left(\frac{4}{3}\right)$ **B.** $r = \sqrt{7}, \quad \theta = \tan^{-1}\left(\frac{3}{4}\right)$

C. $r = 5, \quad \theta = \tan^{-1}\left(\frac{3}{4}\right)$ **D.** $r = 5, \quad \theta = \tan^{-1}\left(\frac{4}{3}\right)$

Q5.15.1 MCQ 2015 1M

A complex-valued function, $f(z)$, given below is analytic in domain D:

$$f(z) = u(x, y) + iv(x, y) \qquad z = x + iy$$

Which of the following is NOT correct?

A. $\frac{df}{dz} = \frac{\partial v}{\partial y} + i\frac{\partial u}{\partial y}$ **B.** $\frac{df}{dz} = \frac{\partial u}{\partial x} + i\frac{\partial v}{\partial x}$

C. $\frac{df}{dz} = \frac{\partial v}{\partial y} - i\frac{\partial u}{\partial y}$ **D.** $\frac{df}{dz} = \frac{\partial v}{\partial y} + i\frac{\partial v}{\partial x}$

Q5.15.2 NAT 2015 2M

For complex variable z, the value of the contour integral $\frac{1}{2\pi i}\oint_C \frac{e^{-2z}}{z(z-3)}dz$ along the clockwise contour $C : |z| = 2$ (up to two decimal places) is _____.

Q5.14.1 MCQ 2014 1M

If $f^*(x)$ is the complex conjugate of $f(x) = \cos(x) + i\sin(x)$, then for real a and b,

$$\int_a^b f^*(x)f(x)\,dx \text{ is } \textbf{ALWAYS}$$

A. positive **B.** negative **C.** real **D.** imaginary

Q5.12.1 MCQ 2012 2M

If $i = \sqrt{-1}$, the value of the integral

$$\oint_c \frac{7z + i}{z(z^2 + 1)}\,dz \qquad |z| < 2,$$

using the Cauchy residue theorem is

A. $2\pi i$ **B.** 0 **C.** -6π **D.** 6π

Q5.08.1 MCQ 2008 2M

An analytic function $w(z)$ is defined as $w = u + iv$, where $i = \sqrt{-1}$ and $z = x + iy$. If the real part is given by $u = \frac{y}{x^2 + y^2}$, $w(z)$ is

A. $\frac{1}{z}$ **B.** $\frac{1}{z^2}$ **C.** $\frac{i}{z}$ **D.** $\frac{1}{iz}$

Q5.07.1 MCQ 2007 2M

If $z = x + iy$ is a complex number, where $i = \sqrt{-1}$ then the derivative of $z\bar{z}$ at $2 + i$ is

A. 0 **B.** 2 **C.** 4 **D.** does not exist

Production & Industrial Engineering

Q5.26.1 MCQ 2026 1M

Which ONE of the following is the value of $\frac{1}{i^n}$? where $i = \sqrt{-1}$, and n is an even positive integer.

A. +1 or −1 **B.** $+i$ or $-i$

C. only +1 **D.** only $-i$

Q5.25.1 MCQ 2025 1M

The eigenvalues of the matrix $\begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix}$ are

A. $-\sqrt{-1}$ and $\sqrt{-1}$ **B.** −1 and 1

C. $1 + \sqrt{-1}$ and $1 - \sqrt{-1}$ **D.** $-1 + \sqrt{-1}$ and $-1 - \sqrt{-1}$

Q5.25.2 MCQ 2025 2M

Which one of the following functions is analytic, given $i = \sqrt{-1}$?

A. $e^x(\cos y + i\sin y)$ **B.** $e^x(\cos y - i\sin y)$

C. $e^x(-\cos y + i\sin y)$ **D.** $e^{-x}(\cos y + i\sin y)$

Q5.24.1 MSQ 2024 1M

In a complex function

$$f(x, y) = u(x, y) + i\,v(x, y),$$

i is the imaginary unit, and $x, y, u(x, y)$ and $v(x, y)$ are real.If $f(x, y)$ is analytic then which of the following equations is/are TRUE?

☐ **A.** $\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} = 0$

☐ **B.** $\frac{\partial^2 v}{\partial x^2} + \frac{\partial^2 v}{\partial y^2} = 0$

☐ **C.** $\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 v}{\partial y^2} = 0$

☐ **D.** $\left(\frac{\partial u}{\partial x}\right)\left(\frac{\partial v}{\partial x}\right) + \left(\frac{\partial u}{\partial y}\right)\left(\frac{\partial v}{\partial y}\right) = 0$

Q5.21.1 NAT 2021 1M

If $(3i + 1)x + (4i + 4)y + 5 = 0$ with x, y being real and $i = \sqrt{-1}$, then $x =$ _____. (correct up to one decimal place)

Q5.20.1 MCQ 2020 1M

For the complex numbers $z_1 = 2 + 3i$ and $z_2 = 4 - 5i$, the value of $(z_1 + z_2)^2$ is

A. $32 - 24i$ **B.** $-32 - 24i$

C. $32 + 24i$ **D.** $-32 + 24i$

Q5.18.1 MCQ 2018 2M

Consider the analytic function $f(z) = x^2 - y^2 + i2xy$ of the complex variable $z = x + iy$, where $i = \sqrt{-1}$. The derivative $f'(z)$ is

A. $2x + i2y$ **B.** $x^2 + iy^2$

C. $x + iy$ **D.** $2x - i2y$

Q5.17.1 MCQ 2017 1M

The product of a complex number $z = x+iy$ and its complex conjugate $\bar{z}$ is

A. x^2 **B.** y^2 **C.** $x^2 - y^2$ **D.** $x^2 + y^2$

Q5.16.1 MCQ 2016 1M

The function

$$f(z) = \frac{z^2 + 1}{z^2 + 4}$$

is singular at

A. $z = \pm 2$ **B.** $z = \pm 1$ **C.** $z = \pm i$ **D.** $z = \pm 2i$

Q5.13.1 MCQ 2013 1M

The eigenvalues of a symmetric matrix are all

A. complex with non-zero positive imaginary part

B. complex with non-zero negative imaginary part

C. real

D. pure imaginary

Q5.11.1 MCQ 2011 2M

The value of $\oint_C \frac{z^2}{z^4 - 1}\,dz$ using Cauchy's integral around the circle $|z + 1| = 1$, where $z = x + iy$, is

A. $2\pi i$ **B.** $-\frac{\pi i}{2}$ **C.** $-\frac{3\pi i}{2}$ **D.** $\pi^2 i$

Q5.10.1 MCQ 2010 1M

If a complex number w satisfies the equation $w^3 = 1$ then the value of $1 + w + \frac{1}{w}$ is

A. 0 **B.** 1 **C.** 2 **D.** 4

Q5.08.1 MCQ 2008 1M

The value of the expression $\frac{-5 + 10\mathrm{i}}{3 + 4\mathrm{i}}$ is

A. $1 - 2\mathrm{i}$ **B.** $1 + 2\mathrm{i}$

C. $2 - \mathrm{i}$ **D.** $2 + \mathrm{i}$

Q5.07.1 MCQ 2007 1M

If a complex variable $z = \frac{\sqrt{3}}{2} + \mathrm{i}\frac{1}{2}$ then z^4 is

A. $2\sqrt{3} + 2\mathrm{i}$ **B.** $\frac{-1}{2} + \mathrm{i}\frac{\sqrt{3}}{2}$

C. $\frac{\sqrt{3}}{2} - \mathrm{i}\frac{1}{2}$ **D.** $\frac{\sqrt{3}}{8} + \mathrm{i}\frac{1}{8}$

Q5.05.5 MCQ 2005 2M ○○○

The function

$$w = u + \mathrm{i}v = \frac{1}{2}\log(x^2 + y^2) + \mathrm{i}\tan^{-1}(y/x)$$

is not analytic at the point

A. $(0, 0)$ **B.** $(0, 1)$ **C.** $(1, 0)$ **D.** $(2, 0)$

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Answer Key

Note: * descriptive/evidence-based, no single definitive answer.

ELECTRONICS & COMMUNICATION ENGINEERING					
Q5.25.1 A,B	Q5.24.1 D	Q5.23.1 A	Q5.23.2 B	Q5.22.1 0.5	Q5.21.1 *
Q5.19.1 D	Q5.19.2 0	Q5.18.1 2	Q5.17.1 B	Q5.17.2 D	Q5.16.1 -136 to -132
Q5.16.2 0	Q5.16.3 B	Q5.15.1 D	Q5.15.2 B	Q5.15.3 9.9 to 10.1	Q5.15.4 0.5
Q5.15.5 B	Q5.14.1 B	Q5.14.2 B	Q5.14.3 -0.6 to -0.4	Q5.12.1 C	Q5.11.1 A
Q5.10.1 C	Q5.09.1 D	Q5.08.1 D	Q5.08.2 A	Q5.07.1 A	Q5.06.1 B
Q5.06.2 D					
ELECTRICAL ENGINEERING					
Q5.26.1 25 to 25.6	Q5.25.1 0	Q5.24.1 D	Q5.24.2 0	Q5.21.1 D	Q5.21.2 C
Q5.20.1 C	Q5.19.1 B	Q5.19.2 A	Q5.18.1 A	Q5.18.2 B	Q5.17.1 D
Q5.17.2 B	Q5.16.1 B	Q5.16.2 B	Q5.15.1 B	Q5.14.1 C	Q5.14.2 B
Q5.14.3 C	Q5.13.1 B	Q5.13.2 A	Q5.12.1 A	Q5.11.1 D	Q5.11.2 D
Q5.08.1 D	Q5.07.1 B	Q5.97.1 B			
INSTRUMENTATION ENGINEERING					
Q5.26.1 A	Q5.25.1 2	Q5.25.2 A,B	Q5.24.1 B	Q5.24.2 A,C	Q5.23.1 A
Q5.22.1 A,C	Q5.22.2 0.49 to 0.51	Q5.21.1 0	Q5.21.2 B	Q5.20.1 B	Q5.19.1 C
Q5.18.1 B	Q5.17.1 B	Q5.16.1 1	Q5.15.1 B	Q5.13.1 D	Q5.11.1 C
Q5.10.1 D	Q5.09.1 D	Q5.09.2 A	Q5.09.3 B	Q5.08.1 B	Q5.07.1 B
Q5.05.1 A	Q5.05.2 C	Q5.02.1 A	Q5.97.1 B	Q5.94.1 C	Q5.94.2 D
MECHANICAL ENGINEERING					
Q5.26.1 A	Q5.25.1 -0.1 to 0.1	Q5.24.1 D	Q5.23.1 1.999 to 2.001	Q5.22.1 A	Q5.22.2 0.2

Q5.21.1 C	Q5.21.2 B	Q5.20.1 D	Q5.20.2 1.99 to 2.01	Q5.20.3 D	Q5.19.1 B
Q5.18.1 A	Q5.17.1 B	Q5.16.1 A	Q5.16.2 -1.1 to -0.9	Q5.16.3 B	Q5.15.1 A
Q5.14.1 C	Q5.14.2 C	Q5.11.1 A	Q5.10.1 B	Q5.09.1 C	Q5.08.1 A
Q5.07.1 B	Q5.06.1 A				
CIVIL ENGINEERING					
Q5.11.1 D	Q5.10.1 D	Q5.09.1 D	Q5.09.2 C	Q5.07.1 A	Q5.06.1 A
Q5.05.1 C	Q5.05.2 A				
CHEMICAL ENGINEERING					
Q5.26.1 D	Q5.25.1 B	Q5.24.1 B	Q5.23.1 B	Q5.20.1 C	Q5.19.1 A
Q5.17.1 2.99 to 3.01	Q5.16.1 D	Q5.15.1 A	Q5.15.2 0.31 to 0.35	Q5.14.1 C	Q5.12.1 B
Q5.08.1 C	Q5.07.1 D				
PRODUCTION & INDUSTRIAL ENGINEERING					
Q5.26.1 A	Q5.25.1 A	Q5.25.2 A	Q5.24.1 A,B,D	Q5.21.1 2.4 to 2.6	Q5.20.1 A
Q5.18.1 A	Q5.17.1 D	Q5.16.1 D	Q5.13.1 C	Q5.11.1 B	Q5.10.1 A
Q5.08.1 B	Q5.07.1 B	Q5.05.5 A			

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PowerX	Electrical Machines, Power Systems, Power Electronics, Measurements and Electromagnetic Field Theory — EE, IN
Engineering Mathematics	Linear Algebra, Calculus, Differential Equations, Complex Variables, Probability & Statistics, Numerical Methods, Transform Theory and Vector Calculus — all branches
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